

1. Purpose & Scope

What we are building

A pre-launch community platform for cancer patients, , and verified medical professionals.
Goal: onboard real users before the full CereOnco product launches, build an engaged audience, and create social proof.
This is V1 — community only. Patient portal, clinical tools, and radiology features come in later phases.
Cognie AI is included in V1 in two subtle ways: (1) Cognie-powered onboarding conversation, (2) Cognie inline community voices inside the chat interface.

2. Tech Stack & Environment

2.1 Confirmed Stack

Layer	Technology
Frontend	React (Vite recommended for fast dev server)
Backend	Python (FastAPI preferred) or Node.js (Express) — TBD with Sayak
Database	PostgreSQL
Hosting	Oracle Cloud Infrastructure (OCI)
Authentication	TBD — evaluate Firebase Auth, Auth0, or custom JWT depending on user volume and OCI config
File Storage	OCI Object Storage (for profile photos, post images)
AI / Cognie	Anthropic Claude API (claude-sonnet-4-20250514) — API key provided separately
Real-time	WebSockets or Server-Sent Events for notifications and online presence

3. User Roles & Permissions

V1 has four user types. Each has a different onboarding path and different permissions within the community.

Role	Who they are	What they can do
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Patient	Someone with a cancer diagnosis. The primary user.	Post, comment, like, join groups, DM, access all community content, use Cognie
Family members/loved ones	Family member or close supporter of a patient.	Same as patient. Can identify themselves in posts as a caregiver.
Medical Professional	Oncologist, radiologist, nurse, psychologist, researcher.	All patient permissions + Verified MD badge on posts and comments. Cannot self-verify — admin approves.
Admin	Internal Cerebree/CereOnco team.	All permissions + user management, content moderation, verification approvals, analytics dashboard.

Verification flow for Medical Professionals

User signs up, selects 'Medical Professional' during Cognie onboarding.
Prompted to upload credentials (medical license number, hospital affiliation).
Status shows 'Pending Verification' — they can use the platform but without the Dr. badge.
Dr can verify automatically if they make account with their affiliated email id and if not - Admin
reviews and approves in the admin dashboard.
Once approved, all their posts and comments display the green 'Verified Dr.' badge.
This is a manual process in V1 — can be automated in V2.

4. Screens & User Workflow

This section maps every screen in V1, what triggers it, and what clicking each element does.
Build in this exact sequence.

4.1 Entry Points

Screen 1 — Landing / Splash

URL: /

What the user sees: CereOnco Community branding, a brief description, and two buttons —
'Join the Community' and 'Sign In'.

Interactions:

- Click 'Join the Community' → navigates to /onboarding (Cognie onboarding flow)
- Click 'Sign In' → navigates to /login
- Any existing session → auto-redirect to /community/feed

Screen 2 — Cognie Onboarding (replaces signup form)

URL: /onboarding

This is the most important and unique screen in V1. Instead of a traditional signup form, Cognie has a conversation with the new user. The conversation collects everything needed to build their profile and personalise their experience.

Conversation flow (Cognie asks in this sequence):

1. Welcome message + ask: 'Are you joining as a patient, a caregiver, or a medical professional?'
2. Based on answer — if patient or caregiver: 'Which cancer are you or your loved one navigating?' (dropdown options + 'Other')
3. 'What stage are you at right now?' (options: just diagnosed / in treatment / between treatments / post-treatment / prefer not to say)
4. 'What matters most to you right now?' (multi-select: peer connection / medical questions / survivor stories / emotional support / research updates)
5. If Medical Professional: 'What is your speciality and where do you practice?' (free text)
6. 'What should we call you?' (first name — free text input)
7. Cognie summary message: 'Welcome, [name]! I've set up your profile and found groups and people that match your journey. Your community is ready.'
8. Auto-redirect to /community/feed after 2 seconds

What happens in the background during onboarding:

- User account is created in the database at Step 1 (email + password collected before conversation starts, on a pre-conversation screen)
- Profile fields populate as conversation progresses
- Group memberships auto-assigned based on condition selected
- Feed personalisation tags set based on 'what matters most' answer
- If Medical Professional selected — account flagged as 'pending verification'

Screen 3 — Login

URL: /login

Standard email + password form. 'Forgot password' link. On success → redirect to /community/feed.

4.2 Main Community Screens

Screen 4 — Community Feed (Home)

URL: /community/feed

The main screen. This is where users spend most of their time.

Layout: three columns — left sidebar (profile + my groups + online now), center feed, right sidebar (stats + trending + suggestions).

Top navigation bar — always visible, sticky:

- CereOnco logo (left) → clicking navigates to /community/feed
- Nav tabs: Home, Groups, Stories, Ask a Doctor, Research, Events
- Search bar → filters posts in real time as user types
- Messages icon with unread badge → opens DM Panel (slide-in drawer)
- User avatar → opens profile dropdown (view profile / settings / logout)

Post Composer (top of center feed):

- Textarea — user types their post
- '✦ Cognie can improve this' button appears when user has typed 10+ characters → calls Cognie API with the draft, returns improved version, user can accept or keep original
- Emoji, photo, tag buttons (photo upload goes to OCI Object Storage in V1)
- Post button → creates post via API, inserts at top of feed

Post Card (each post in the feed):

- Avatar + name + role + verified badge (if MD) + post type tag + timestamp
- Post content (text, image if present)
- Like button → optimistic UI update + API call
- Comment button + count → opens Comment Drawer (slide-in from right)
- Share button → copies URL to clipboard
- Save/bookmark button → saves to user's saved posts

Screen 5 — Comment Drawer

Trigger: clicking comment button on any post

UI: slides in from the right side (not a full page — overlay drawer)

Contains: original post preview at top, threaded comments below, reply composer at bottom.

Reply composer interactions:

- User types reply
- '✦ Cognie help' button appears → calls Cognie API to gently improve the reply, user can accept or keep their version
- Submit → POST /api/comments → reply appears immediately

Screen 6 — Direct Messages

Trigger: clicking the messages icon in the top nav

UI: slide-in panel from the right (not a full page)

Contains: conversation list on left, active conversation on right.

Interactions:

- Click a conversation → opens that thread
- Type + send message → POST /api/messages → appears immediately
- '+ New Message' button → search for a user by name, start a new thread
- Unread badge on nav icon updates in real time

Screen 7 — Groups

URL: /community/groups

Grid of all available groups. Each group card shows: icon, name, member count, activity status, short description, Join/Joined button.

Interactions:

- Join button → POST /api/groups/:id/join → button changes to 'Joined ✓', group appears in left sidebar
- Clicking a group card → navigates to /community/groups/:id (group feed — same structure as main feed but filtered to that group)

Screen 8 — Survivor Stories

URL: /community/stories

Full-page list of long-form survivor stories. Each card shows: author, headline, preview text, read time, condition, reaction count, 'Read story' button.

Interactions:

- 'Read story' → navigates to /community/stories/:id (full story page)
- 'Write your story' → opens a story composer modal (rich text, longer format than a post)

Screen 9 — Ask a Doctor (will send another doc specifically for this)

URL: /community/ask

Top section: Cognie AI panel — user asks any cancer question, gets an instant response.
Below: posts from verified MDs displayed in a separate feed.

Cognie panel interactions:

- User types question + clicks 'Ask' or presses Enter
- Loading state (Cognie typing indicator)
- Answer appears with clinical response
- Quick suggestion chips below input (pre-filled questions)

Screen 10 — Research

URL: /community/research

Curated research updates. Cards with: icon, source, date, badge (New/Featured), title, summary, 'Read summary' button.

V1: content is manually curated by admin. V2 can add a feed integration.

Screen 11 — Events

URL: /community/events

Grid of upcoming events and webinars. Each card shows: date, title, host, event type badge, attendee count, RSVP button.

'RSVP' → records interest in database + shows confirmation.

Screen 12 — User Profile

URL: /community/profile/:userId

Shows: avatar, name, role, condition, join date, post count, post history, mutual connections.

Own profile has an Edit button. Other profiles have a Follow / Message button.

Screen 13 — Admin Dashboard

URL: /admin (separate route, admin role only — redirect to /login if not admin)

Tabs: Users, Content Moderation, Verification Queue, Analytics.

Users tab:

- Table of all users with role, join date, post count, status
- Actions: suspend, delete, change role

Verification Queue tab:

- List of Medical Professional accounts awaiting verification
- Each row shows: name, specialty, hospital, uploaded credentials
- Actions: Approve (adds MD badge) or Reject (sends notification)

Content Moderation tab:

- Reported posts and comments
- Actions: dismiss report, remove content, warn user

Analytics tab:

- Total users, DAU, posts per day, top groups, top posts
- V1 can be simple counts — no complex charts needed

4.3 Cognie Floating Button (Community-wide)

A subtle floating action button (✚) is always visible in the bottom-right corner of all community screens.

Clicking it opens the Cognie Chat Modal — a full chat interface that stays on top of whatever screen the user is on.

The Cognie Chat Modal automatically surfaces relevant community voices when the user asks questions that match known community experiences (e.g. asking about fatigue during chemo shows 2-3 real community posts from people who had the same experience, inline in Cognie's response).

5. API Endpoints

RESTful API. All endpoints except auth require a valid JWT token in the Authorization header.

Method + URL	Description	Auth required
POST /api/auth/register	Create account (called at start of onboarding)	No
POST /api/auth/login	Email + password login, returns JWT	No
POST /api/auth/logout	Invalidate session	Yes
GET /api/users/me	Get current user profile	Yes
PATCH /api/users/me	Update profile	Yes
GET /api/users/:id	Get any user's public profile	Yes
GET /api/posts	Get feed posts (paginated, filtered by tags)	Yes
POST /api/posts	Create a post	Yes
DELETE /api/posts/:id	Delete own post	Yes
POST /api/posts/:id/like	Like a post	Yes
DELETE /api/posts/:id/like	Unlike a post	Yes
GET /api/posts/:id/comments	Get comments for a post	Yes
POST /api/posts/:id/comments	Add a comment	Yes
GET /api/groups	Get all groups	Yes
POST /api/groups/:id/join	Join a group	Yes
DELETE /api/groups/:id/join	Leave a group	Yes
GET /api/messages	Get all conversations for current user	Yes
GET /api/messages/:userId	Get thread with specific user	Yes

POST /api/messages/:userId	Send a message	Yes
POST /api/cognie/chat	Send message to Cognie AI, returns response + community matches	Yes
POST /api/cognie/improve-post	Cognie improves a post draft	Yes
GET /api/admin/users	List all users — admin only	Admin
PATCH /api/admin/users/:id/verify	Approve MD verification — admin only	Admin

7. Cognie AI Integration

Cognie uses the Anthropic Claude API. Two integration points in V1:

7.1 Cognie Onboarding

The onboarding conversation is frontend-driven in V1. The sequence of questions is hardcoded in the React component. Cognie API is NOT called for each question — only the final welcome message is generated dynamically via the API to personalise it with the user's name and condition.

API call for final onboarding message

Model: claude-sonnet-4-20250514

System: 'You are Cognie, CereOnco's AI companion. Write a warm 2-sentence welcome message for a new user joining the cancer support community. Use their name and acknowledge their specific situation.'

User message: 'Name: [name], Role: [role], Condition: [condition], Stage: [stage]'

Max tokens: 150

7.2 Cognie Chat (floating button)

Full conversational AI. When the user sends a message, the backend:

9. Receives the message and conversation history
10. Searches the posts table for semantically similar posts (simple keyword matching in V1 — semantic search can come in V2)
11. If matching community posts found, includes them as context in the Cognie response
12. Calls Claude API with system prompt + conversation history
13. Returns both the AI response and the matched community posts to the frontend
14. Frontend renders the AI response + inline community voice card (if matches exist)

System prompt for Cognie Chat

You are Cognie, a warm and knowledgeable AI companion for CereOnco — a community platform for cancer patients, caregivers, and medical professionals.
Be compassionate, evidence-based, and personal. Use plain language.
Keep responses to 3-4 sentences.
Always remind users to consult their oncology team for advice specific to their case.
Never diagnose or prescribe.

7.3 Cognie Post Improvement

When a user clicks '💡 Cognie can improve this' while composing a post or comment:

15. Frontend sends the draft text to POST /api/cognie/improve-post
16. Backend calls Claude API with the draft
17. Returns improved version — user can accept or keep original

System prompt: 'You are Cognie helping a cancer patient write a compassionate, authentic community post. Improve their draft slightly — keep their voice, make it warmer and clearer. Return only the improved text.'

8. 2-Week Build Timeline

Day	What gets built	Owner
Day 1–2	Project setup, OCI config, DB schema, auth endpoints (register/login), basic React scaffold with routing	
Day 2–3	Cognie onboarding flow (frontend conversation UI + profile creation API)	
Day 3–5	Community feed — post creation, display, like, comment drawer, post composer with Cognie improve	
Day 4–5	Groups — list, join/leave, group feed. Left sidebar with my groups + online now.	
Day 5–6	DM system — conversation list, threaded messages, unread count	
Day 6–7	Survivor stories, Ask a Doctor (Cognie + MD posts), Research, Events pages	

Day 7	Cognie floating button + chat modal + inline community voice logic	
Day 8–9	Admin dashboard — user management, verification queue, moderation	
Day 9–10	Right sidebar — trending, suggestions, stats. User profile pages. Follow system.	
Day 10–11	Real-time features — online presence, DM notifications, unread badges	
Day 11–12	OCI deployment — set up production environment, deploy backend + frontend, connect DB	
Day 12–13	QA, bug fixes, content seeding (first posts and groups), performance check	Full team
Day 14	Internal review with. Final fixes. Soft launch.	Full team

9. Out of Scope for V1

The following are explicitly NOT part of V1. Do not build these — they will be specified separately.

- Oncopedia / knowledge hub
- Push notifications (email notifications only for V1 if time permits)
- Semantic / vector search for Cognie community matching (use keyword matching in V1)
- Payment or subscription system
- Mobile app (web-responsive only for V1)
- Video calling or live webinar hosting (RSVP only in V1 — no hosting)
- Automated MD verification (manual admin review only in V1)

Important notes for the developer

When in doubt about product decisions — come to Arib same day. Do not assume.
OCI setup — coordinate with Sayak/Agniva. Use OCI Compute for the backend, OCI Object Storage for file uploads, and OCI Autonomous Database or a self-managed PostgreSQL on Compute.

Authentication is TBD — make the auth layer modular so it can be swapped without rebuilding the rest of the app.

Mobile responsiveness is required — design for 375px minimum width.

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